PRACTICE SHEET

1. What does the solution of the differential equation $xdy - ydx = 0$ represent?
(a) Rectangular hyperbola
(b) Straight line passing through (0, 0)
(c) Parabola with vertex at (0, 0)
(d) Circle with centre at (0, 0)

2. Which one of the following differential equations represents the system of circles touching y – axis at the origin?
(a) $\frac{dy}{dx} - x^2 - y^2$ (b) $2xy\frac{dy}{dx} = y^2 - x^2$
(c) $2xy\frac{dy}{dx} = x^2 - y^2$ (d) $\frac{dy}{dx} = y^2 - x^2$

3. What is the solution of the differential equation $\frac{dy}{dx} = \frac{y}{(x+2y^3)}$?
(a) $y(1 - xy) = cx$ (b) $y^3 - x = cy$
(c) $x(1 - xy) = cy$ (d) $x(1 + xy) = cy$

4. If $y^2 = p(x)$ is a polynomial of degree 3, then what is $2\frac{d}{dx}\left[y^3\frac{d^2y}{dx^2}\right]$ equal to?
(a) $p'(x)p'''(x)$ (b) $p''(x)p'''(x)$

9. For what value of k does the differential equation $\frac{dy}{dx} = ky$ represent the law of natural decay?
(a) –5 (b) 0
(c) 0.01 (d) (10)–1

10. What is the solution of the differential equation $(x + y)(dx - dy) = dx + dy$?
(a) $x + y + \ln(x + y) = c$
(b) $x - y + \ln(x + y) = c$
(c) $y - x + \ln(x + y) = c$
(d) $y - x - \ln(x - y) = c$

11. What is the degree of the differential equation, $k\frac{d^2y}{dx^2} = \left[1+\left(\frac{dy}{dx}\right)^3\right]^{3/2}$, where k is a constant?
(a) 1 (b) 2
(c) 3 (d) 4

12. Under which one of the following conditions does the solution of $\frac{dy}{dx} = \frac{ax+b}{cy+d}$ represent a parabola?
(a) $a = 0, c = 0$
(b) $a = 1, b = 2, c \neq 0$
(c) $a = 0, c \neq 0, b \neq 0$